Benchmarking case for Aero-servo-elastic LES codes

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1 Benchmarking Case

- Turbine: IEA 15 MW reference wind turbine, single-rotor configuration.
- Structural configuration: no nacelle and no tower (pure rotor-only loading).
- Purpose: reference benchmark for existing aero-servo-elastic LES codes in a simple uniform inflow.
- OpenFAST input files for this case can be downloaded using the following link :github

2 Case Setup

2.1 Solver and Models

- CFD solver: LES based solver with incompressible Navier-Stokes formulation.
- Wind-turbine model: Actuator Line Model (ALM).
- Mesh type: Cartesian grid.
- Numerical schemes: (for YALES2)
  - Space: 4th-order scheme.
  - Time: TFV4A scheme.

2.2 Geometrical Setup

- Rotor diameter: $D = 240\,\mathrm{m}$.
- Domain size: $18D \times 6D \times 6D$.
- Physical coordinates:
  - $x \in [0, 4320]$ m
  - $y \in [0, 1440]$ m
  - $z \in [0, 1440]$ m
- Turbine configuration: no tower, no nacelle.
- Rotor centre: $(3D, 3D, 150m)$ (i.e. $(720\,\mathrm{m}, 720\,\mathrm{m}, 150\,\mathrm{m})$), rotor located at domain centre; OpenFAST rotor positions are transformed to this reference.
- Number of actuator points per blade: 64.
- Pitch angle: 0°; yaw angle: 0°; cone angle: $-4°$;tilt angle: $-6°$.
- Mollification kernel width: $\varepsilon/\Delta x = 3$.

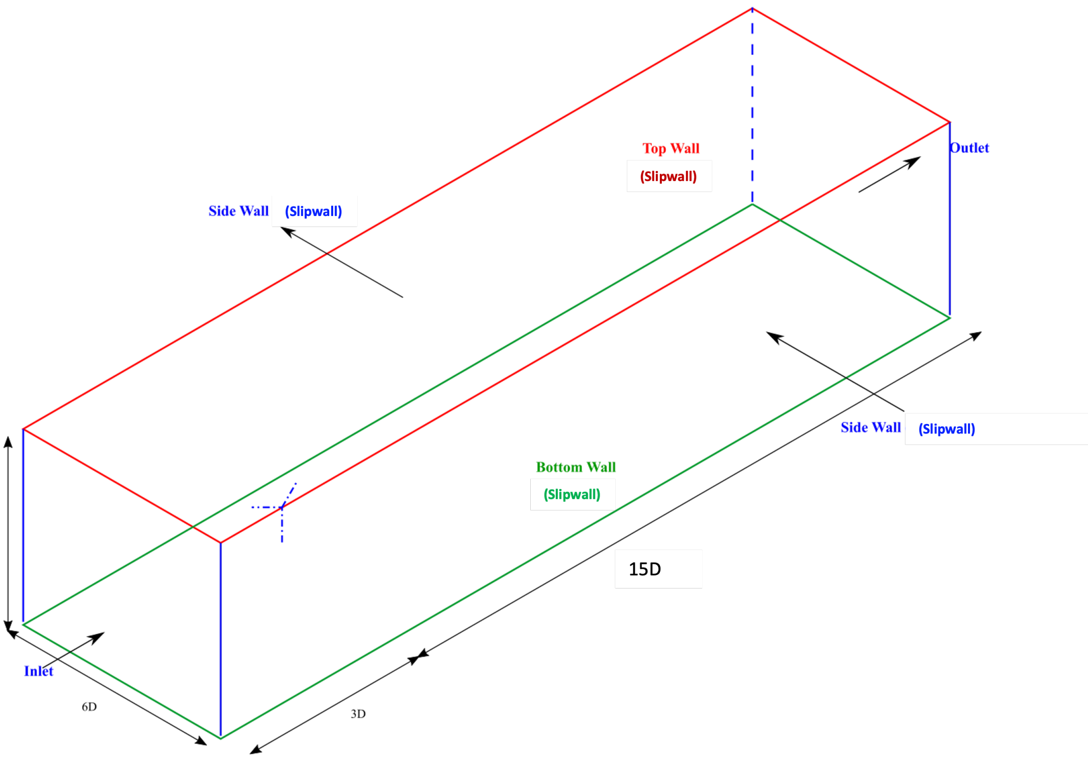

Figure 1: Schematic of computational domain and boundary conditions.

2.3 Mesh and Resolution

- Mesh type: uniform Cartesian grid, no local refinement zones.
- Grid spacing: $\Delta x = \Delta y = \Delta z = 4\,\mathrm{m}$. Initial run with a coarser refinement level $8m$ resolution.
- Non-dimensional resolution: $D/\Delta x = 60$.
- Approximate number of cells:

$$N_{\text{cells}} = \frac{4320}{4} \times \frac{1440}{4} \times \frac{1440}{4} \approx 1.39 \times 10^8 \text{ cells.}$$

3 Inflow and Boundary Conditions

- Inflow: uniform, non-turbulent inflow with

$$U_\infty = 8\,\mathrm{m\,s^{-1}}.$$

- Boundary conditions:
  - Slip walls on all four lateral walls.
  - inlet and outlet (x :streamwise direction) for front and back walls.

4 Turbine Parameters

see Table 1

Quantity	Value
Hub radius	3.97 m
Blade length	116.03 m
Rotor diameter	240 m
Tip-speed ratio (TSR)	9
Rotation speed	$\omega = 0.60\,\mathrm{rad/s}$
Rotation speed	$\approx 5.73\,\mathrm{rpm}$
Cone angle	$-4°$
Controller	off

Table 1: Main turbine parameters used in the benchmark case.

5 Simulation Schedule

- Time step: $\Delta t = 0.025\,\mathrm{s}$.
- Streamwise flow-through time:

$$T_{\mathrm{FT}} = \frac{L_x}{U_\infty} = \frac{18D}{U_\infty} = 540\,\mathrm{s}.$$

- Initial transient discard: one flow-through time, 540 s.
- Statistics accumulation: two flow-through times, $2 \times 540 = 1080\,\mathrm{s}$.
- Total simulated physical time:

$$T_{\mathrm{total}} = 540 + 1080 = 1620\,\mathrm{s}.$$

6 Post-processing and Diagnostics

6.1 Planes for Field Data

- Y-normal plane at rotor centre location: $y = 3D$.
- Z-normal plane at rotor centre location: $z = 3D$.
- X-normal plane at rotor location (rotor plane).

For each plane, compute and compare:

- Mean and RMS velocity components.
- Instantaneous and derived vorticity fields.

These planes are intended for flow field comparison, either ParaView or Python (slices or extracted surfaces).

6.2 Line Probes

Line probes are defined along the rotor centreline (vertical direction through the rotor hub) for a set of downstream streamwise locations. 0D location is considered to be rotor position.

- Streamwise probe locations: $x/D = -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$.
- At each x/D, a vertical line probe along the rotor centreline (through hub height).
- Recorded quantities: mean velocity and RMS velocity components.
- Each line to be divided into 120 points.

6.3 Point Probes

Point probes are defined along the rotor centreline (vertical direction through the rotor hub) for a set of downstream streamwise locations. The idea for point probes is to compare the tip-vortex breakdown phenomena between different codes.

- Streamwise probe locations: $x/D = -2, -1, 0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 6, 7, 8, 9$ at rotor tip locations.
- At each x/D, a point probes along the rotor centerline (through tip height).
- Recorded quantities: time series of instantaneous velocity for either all time steps or atleast 2 flow-through times (time for sampling statistics post transients).

7 Coupling Inputs

- OpenFAST turbine input files (ElastoDyn, AeroDyn and Inflow Wind) are to used
- The input file case-setup will be distributed together with this document.
- The OpenFAST rotor coordinates are transformed such that the rotor centre coincides with $(3D, 3D, 3D)$ in the CFD domain.

8 Summary

This document defines a deterministic benchmark case for existing aero-servo-elastic couplings using the IEA 15 MW reference turbine in uniform, non-turbulent inflow. All geometric parameters, numerical settings, mesh resolution, inflow conditions, simulation schedule, and post-processing diagnostics are specified so that runs can be reproduced and compared across different platforms and coupling implementations.

Following codes are taking part :

- YALES2-OpenFAST : Anand Parinam
- TOSCA-OpenFAST : Sebastiano Stipa
- ASPIRE-OpenFAST : Evert Wiegnant
- AMRWind-OpenFAST : Benito Dello Iacono